

Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds

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September 16, 2026

Why acoustic surveillance

- Shootings in public spaces are more frequent; global gun ownership is rising.
- Growing demand for autonomous security in communal areas — campuses, hospitals, parks, government facilities.
- Cameras, drones and security personnel give *some* protection, but fall short in high-risk situations with an active shooter.

What an acoustic system adds

Identifies gunfire, pinpoints its source, and tracks an ongoing threat — faster detection and shorter response times for security forces.

The acoustic signature of a gunshot

Two components define the pattern:

Muzzle blast

- Explosive gases expanding from the barrel.
- Travels at the speed of sound in the bullet's direction.
- Lasts 1–3 ms, often resembling an “N” shape.

Shock wave (“N-wave”)

- Appears when bullet speed exceeds the speed of sound ($v > c$).
- Produced by compression and expansion shocks.

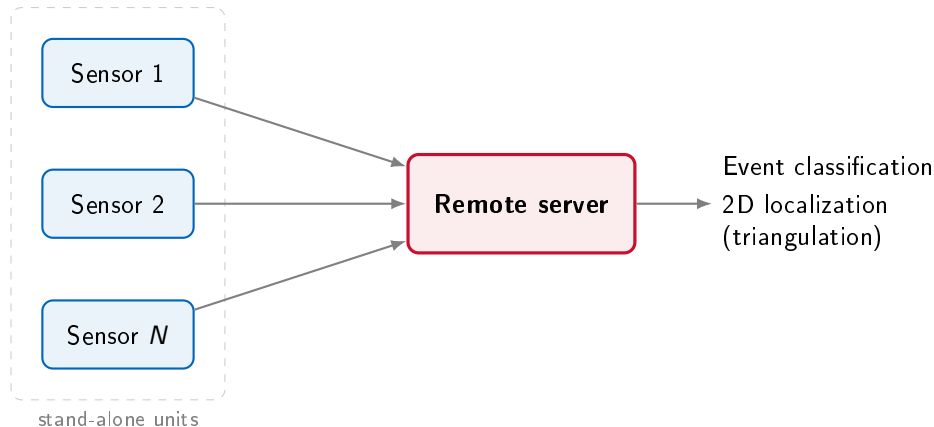
Why naive processing is not enough

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

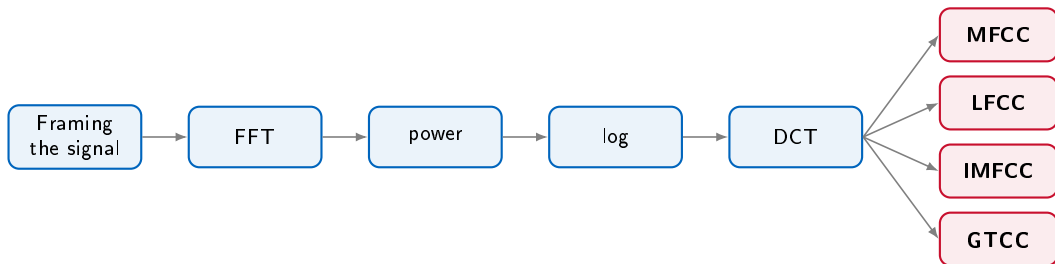
Effective feature extraction must integrate *both* time and frequency domain processing.

Proposed system architecture



- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
- Multiple units give broader coverage and adaptability to environments.
- Tested with various firearm calibers and ammunition.

Feature extraction — cepstral coefficients



Block diagram of the feature extraction algorithms (Fig. 1 in the abstract); each branch applies its own filter bank before the DCT.

- MFCC was developed for other acoustic applications but works for gunshot recognition — validated at military shooting ranges.
- LFCC and IMFCC offer alternative filtering approaches.
- GFCC and GTCC derive from human auditory response models; **GTCC is more noise-resistant than MFCC.**

Classification results — 9 mm class

Frame	Extractor	ACC (%)		RLC (%)		MCC (-)	
		SVM	NN	SVM	NN	SVM	NN
15 ms	MFCC	83,90	85,59	80,65	67,74	0,62	0,62
	LFCC	83,90	83,90	80,65	64,52	0,62	0,57
	IMFCC	85,59	90,68	83,87	87,10	0,66	0,77
	GTCC	93,22	97,46	87,10	93,55	0,82	0,93
30 ms	MFCC	84,75	87,29	67,74	87,10	0,60	0,70
	LFCC	87,29	83,90	70,97	70,97	0,66	0,59
	IMFCC	85,59	92,37	67,74	80,65	0,62	0,80
	GTCC	97,46	98,31	90,32	93,55	0,93	0,96
50 ms	MFCC	84,75	93,22	74,19	90,32	0,61	0,83
	LFCC	88,98	86,44	77,42	77,42	0,71	0,66
	IMFCC	89,83	90,68	74,19	80,65	0,73	0,76
	GTCC	97,46	99,15	90,32	96,77	0,93	0,98

Table I in the abstract —

example of classification results for a 9 mm class.

What the numbers say

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** — better at modelling complex patterns. The gap widens with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms — 99,15 % ACC, 96,77 % RLC, MCC 0,98.

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you — questions?

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Backup — other cepstral coefficients

- **LPCC** (Linear Prediction Cepstral Coefficients) and **PLPC** (Perceptual Linear Prediction Coefficients) — noise elimination, speech recognition.
- **GFCC** (Greenwood Function Cepstral Coefficients) — derived from human auditory response models, used for classifying animal sounds.
- **GTCC** (Gammatone-Frequency Cepstral Coefficients) — same family, greater resistance to noise than MFCC.

Backup — metrics

ACC Accuracy — share of correctly classified events.

RLC Recall — share of actual gunshots that were detected.

MCC Matthews Correlation Coefficient — balanced measure robust to class imbalance; +1 perfect, 0 random.

TODO: confirm these expansions match the definitions used in the paper.